

MATH3303 – Lecture 10

Group Presentations; Classification of Finite Abelian Groups I

Review Notes

Reference: Lecture notes (presentations); Lee, Abstract Algebra, §5.1 (direct products)

Overview

- **Group presentations:** formal definition of $\langle S \mid R \rangle$; proof that the dihedral group D_n has the presentation $\langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle$.
- **Exercise:** presentation of Sym_n via simple reflections and the full set of Coxeter relations.
- **Classification of finite abelian groups:** the internal direct product (Lee, Def. 5.2); exercise on commuting factors (Lee, Lemma 5.1); Lemma 1 (unique expression, Lee, Lemma 5.2).

1 Group Presentations

Definition 1.1: Group presentation (from lecture)

Let S be a set and $R \subseteq F(S)$ (where $F(S)$ is the free group on S). The *presentation* $\langle S \mid R \rangle$ defines the group

$$\langle S \mid R \rangle := F(S) / \langle\langle R \rangle\rangle,$$

where $\langle\langle R \rangle\rangle$ is the normal closure of R in $F(S)$ (the smallest normal subgroup of $F(S)$ containing R , from Lecture 9).

Elements of S are called *generators*; elements of R are called *relators*. A *relation* is an equation $w = 1$ for $w \in R$.

Remark 1.1: Intuition (from lecture)

The group $\langle S \mid R \rangle$ is the “freest possible” group on generators S subject to the constraints R . More precisely: any group G generated by elements satisfying the relations R is a quotient of $\langle S \mid R \rangle$.

Example 1.1: The dihedral group D_n (from lecture)

For $n \geq 2$, the dihedral group D_n (the group of symmetries of a regular n -gon, of order $2n$) has presentation

$$D_n = \langle r, s \mid r^n = 1, s^2 = 1, srs^{-1} = r^{-1} \rangle.$$

Geometrically: r is rotation by $\frac{2\pi}{n}$, s is a reflection, and the relation $srs^{-1} = r^{-1}$ (equivalently $srs = r^{-1}$ since $s^2 = 1$) says that conjugation by s inverts r .

We verify this by constructing an isomorphism $\psi : \langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle \rightarrow D_n$.

Proof that the presented group is isomorphic to D_n . Let $P := \langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle$.

Define $\psi : P \rightarrow D_n$ by sending the generator r to the rotation $R := R_{2\pi/n}$ and the generator s to a fixed reflection S of the n -gon:

$$\psi(r) = R, \quad \psi(s) = S.$$

Step 1: ψ is well-defined.

Since P is a quotient of the free group on $\{r, s\}$, specifying ψ on generators extends to a homomorphism $F(\{r, s\}) \rightarrow D_n$. We must check that the image of each relator is trivial:

- $\psi(r^n) = R^n = \text{id}$ (rotation by 2π is the identity). ✓
- $\psi(s^2) = S^2 = \text{id}$ (a reflection has order 2). ✓
- $\psi(srs) = SRS$ and $\psi(r^{-1}) = R^{-1}$. In D_n , conjugating a rotation by a reflection gives the inverse rotation: $SRS = R^{-1}$. ✓

Hence the relations in P hold in D_n , so ψ factors through P and is a well-defined homomorphism.

Step 2: ψ is surjective.

D_n is generated by R and S , and both lie in the image of ψ .

Step 3: elements of P have a normal form.

In P , we can always rewrite by moving s to the right using $srs = r^{-1}$, i.e. $sr = r^{-1}s$. More generally, $sr^i = r^{-i}s$ for all $i \in \mathbb{Z}$.

Iterating: every word in $\{r, s\}$ and their inverses can be rewritten as

$$r^i \quad \text{or} \quad r^i s, \quad 0 \leq i \leq n-1.$$

(Using $r^n = 1$ and $s^2 = 1$ to reduce exponents.)

Hence $|P| \leq 2n$.

Step 4: ψ is an isomorphism.

$|D_n| = 2n$ and $|P| \leq 2n$. Since ψ is surjective, $|P| \geq |D_n| = 2n$. Hence $|P| = 2n$ and ψ is a bijection.

Therefore $P \cong D_n$. □

Exercise 1.1: Presentation of Sym_n (from lecture)

Let $s_i = (i \ i+1)$ be simple reflections in Sym_n (Lecture 9). Show that

$$\text{Sym}_n \cong \langle s_1, s_2, \dots, s_{n-1} \mid s_i^2 = 1, (s_i s_{i+1})^3 = 1, (s_i s_j)^2 = 1 \text{ for } |i - j| \geq 2 \rangle.$$

The three families of relations are:

- (1) **Order-2 relations:** $s_i^2 = 1$ for all i .
- (2) **Braid relations:** $s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$ for $1 \leq i \leq n-2$.
- (3) **Far-commutativity:** $s_i s_j = s_j s_i$ for $|i - j| \geq 2$.

(This is the *Coxeter presentation* of Sym_n .)

2 Classification of Finite Abelian Groups: Internal Direct Products

2.1 Definition and Commuting Lemma

Definition 2.1: Internal direct product (from lecture; cf. Lee, Def. 5.2)

Let G be a group and let $N_1, N_2, \dots, N_k$ be subgroups of G . We say G is the *internal direct product* of $N_1, N_2, \dots, N_k$ if:

- (1) Each $N_i \trianglelefteq G$;
- (2) $G = N_1 N_2 \cdots N_k$; and
- (3) $(N_1 N_2 \cdots N_i) \cap N_{i+1} = \{1\}$ for $1 \leq i < k$.

Exercise 2.1: Commuting factors (from lecture; cf. Lee, Lemma 5.1)

Let G be a group with normal subgroups $N, K \trianglelefteq G$. If $N \cap K = \{1\}$, then $kn = nk$ for all $k \in K$ and $n \in N$.

Hint. Consider the commutator $k^{-1}n^{-1}kn$, and show it lies in both K and N .

Proof. Let $h = k^{-1}n^{-1}kn$.

$h \in K$: Since $K \trianglelefteq G$, $n^{-1}kn \in K$. So $h = k^{-1}(n^{-1}kn) \in K$.

$h \in N$: Since $N \trianglelefteq G$, $k^{-1}nk \in N$. Write $h = (k^{-1}n^{-1}k)n$... more cleanly: $h = k^{-1}(n^{-1}kn)$ can also be rearranged as $h = (k^{-1}n^{-1}k)n$. Since $N \trianglelefteq G$, $k^{-1}n^{-1}k \in N$, so $h = (k^{-1}n^{-1}k) \cdot n \in N$.

Since $h \in K \cap N = \{1\}$, we have $k^{-1}n^{-1}kn = 1$, i.e. $kn = nk$. $\square$

2.2 Lemma 1: Unique Expression

Lemma 2.1: Unique expression in an internal direct product (from lecture; cf. Lee, Lemma 5.2)

If G is the internal direct product of $N_1, N_2, \dots, N_k$, then every element of G can be written *uniquely* as $n_1 n_2 \cdots n_k$ with each $n_i \in N_i$.

Proof. Existence. Since $G = N_1 N_2 \cdots N_k$, every element of G can be written in such a way.

Uniqueness. We use induction on k .

Base case $k = 1$: Nothing to prove.

Inductive step: Assume the result holds for internal direct products of $k - 1$ subgroups, and suppose

$$n_1 n_2 \cdots n_{k-1} n_k = h_1 h_2 \cdots h_{k-1} h_k, \quad n_i, h_i \in N_i.$$

Then

$$h_k n_k^{-1} = (h_1 \cdots h_{k-1})^{-1} (n_1 \cdots n_{k-1}) \in N_k \cap (N_1 N_2 \cdots N_{k-1}) = \{1\}.$$

Hence $n_k = h_k$, and cancelling gives

$$n_1 n_2 \cdots n_{k-1} = h_1 h_2 \cdots h_{k-1} \quad \text{in } N_1 N_2 \cdots N_{k-1},$$

which is an internal direct product of $k - 1$ subgroups. By the inductive hypothesis, $n_i = h_i$ for all $i \leq k - 1$.

So the expression is unique. □

Theorem 2.1: Internal $\cong$ external direct product (from lecture; cf. Lee, Thm. 5.2)

Let G be a group that is the internal direct product of $N_1, N_2, \dots, N_k$. Then

$$G \cong N_1 \times N_2 \times \cdots \times N_k.$$

Remark 2.1: Convention (from lecture; cf. Lee)

By Theorem 2.3, we may write $G = N_1 \times N_2 \times \cdots \times N_k$ for both internal and external direct products without ambiguity. Context determines which is meant.

Checklist of Skills

- ☐ State the definition of a group presentation $\langle S \mid R \rangle = F(S)/\langle\langle R \rangle\rangle$.
- ☐ Write the presentation of D_n and identify the generators and relations geometrically.
- ☐ Prove that $\langle r, s \mid r^n = 1, s^2 = 1, srs = r^{-1} \rangle \cong D_n$: check well-definedness (relations hold in D_n), surjectivity, normal form (move s to the right using $sr = r^{-1}s$), and element count.
- ☐ State the three families of relations in the Coxeter presentation of Sym_n .
- ☐ State the definition of an internal direct product (three conditions).
- ☐ Prove the commuting lemma: if $N \cap K = \{1\}$ and both are normal, then $NK = KN$ elementwise (via commutator argument).
- ☐ State and prove Lemma 1 (unique expression) by induction on k , using the intersection condition $N_k \cap (N_1 \cdots N_{k-1}) = \{1\}$.
- ☐ State the theorem: internal direct product $\cong$ external direct product.